

LAB 3 – EL1010

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Assignment 1: Calculate the transfer function

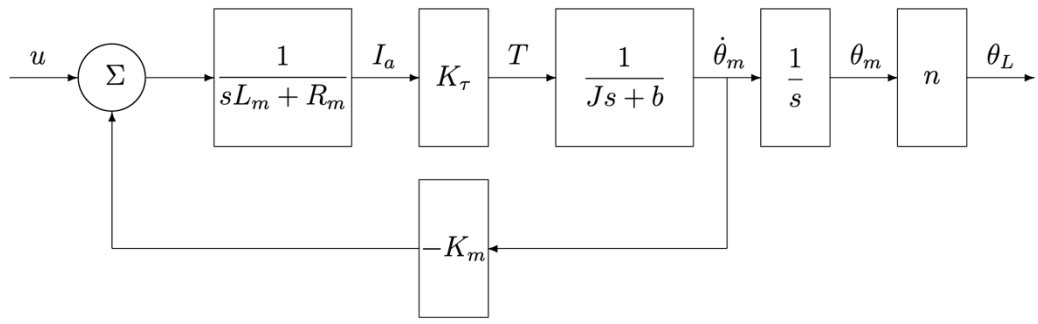
$$G(s) = \frac{1}{s} \frac{n \cdot K_T}{(Js+b)(sL_m+R_m) + K_T K_m} =$$

$$= \frac{1}{s} \frac{\frac{1}{20} \cdot 38}{(5s+1)(2s+21) + 38 \cdot 0,5} =$$

$$= \frac{1,9}{10s^3 + 107s^2 + 40s}$$

G =

$$\frac{1.9}{10 s^3 + 107 s^2 + 40 s}$$



Transfer function $G(s) = \frac{\theta_L}{u}$

$$\theta_L = n \theta_m = n \frac{1}{s} \dot{\theta}_m \quad *$$

$$\theta_m = \frac{1}{s} \dot{\theta}_m$$

$$\dot{\theta}_m = \frac{1}{Js+b} T$$

$$T = K_T I_a, \quad I_a = \frac{1}{sL_m+R_m} \cdot E = \frac{1}{sL_m+R_m} (u - K_m \dot{\theta}_m)$$

$$\Rightarrow \dot{\theta}_m = \frac{1}{Js+b} K_T \frac{1}{sL_m+R_m} (u - K_m \dot{\theta}_m)$$

$$\Rightarrow \dot{\theta}_m \left(1 + \frac{1}{Js+b} K_T \frac{1}{sL_m+R_m} K_m \right) = \frac{1}{Js+b} K_T \frac{1}{sL_m+R_m} u$$

$$\Rightarrow \frac{\dot{\theta}_m}{u} = \frac{\frac{1}{Js+b} K_T \frac{1}{sL_m+R_m}}{1 + \frac{1}{Js+b} K_T \frac{1}{sL_m+R_m} K_m} = \frac{\frac{K_T}{(Js+b)(sL_m+R_m)}}{1 + \frac{K_T K_m}{(Js+b)(sL_m+R_m)}} = \frac{K_T}{(Js+b)(sL_m+R_m) + K_T K_m}$$

$$* \Rightarrow \frac{\dot{\theta}_m}{u} = \frac{\theta_m \cdot s}{u} = \frac{K_T}{(Js+b)(sL_m+R_m) + K_T K_m} \Rightarrow \theta_m = \frac{1}{s} \frac{K_T}{(Js+b)(sL_m+R_m) + K_T K_m} u$$

$$\Rightarrow \frac{\theta_L}{u} = \frac{n}{s} \frac{K_T}{(Js+b)(sL_m+R_m) + K_T K_m}$$

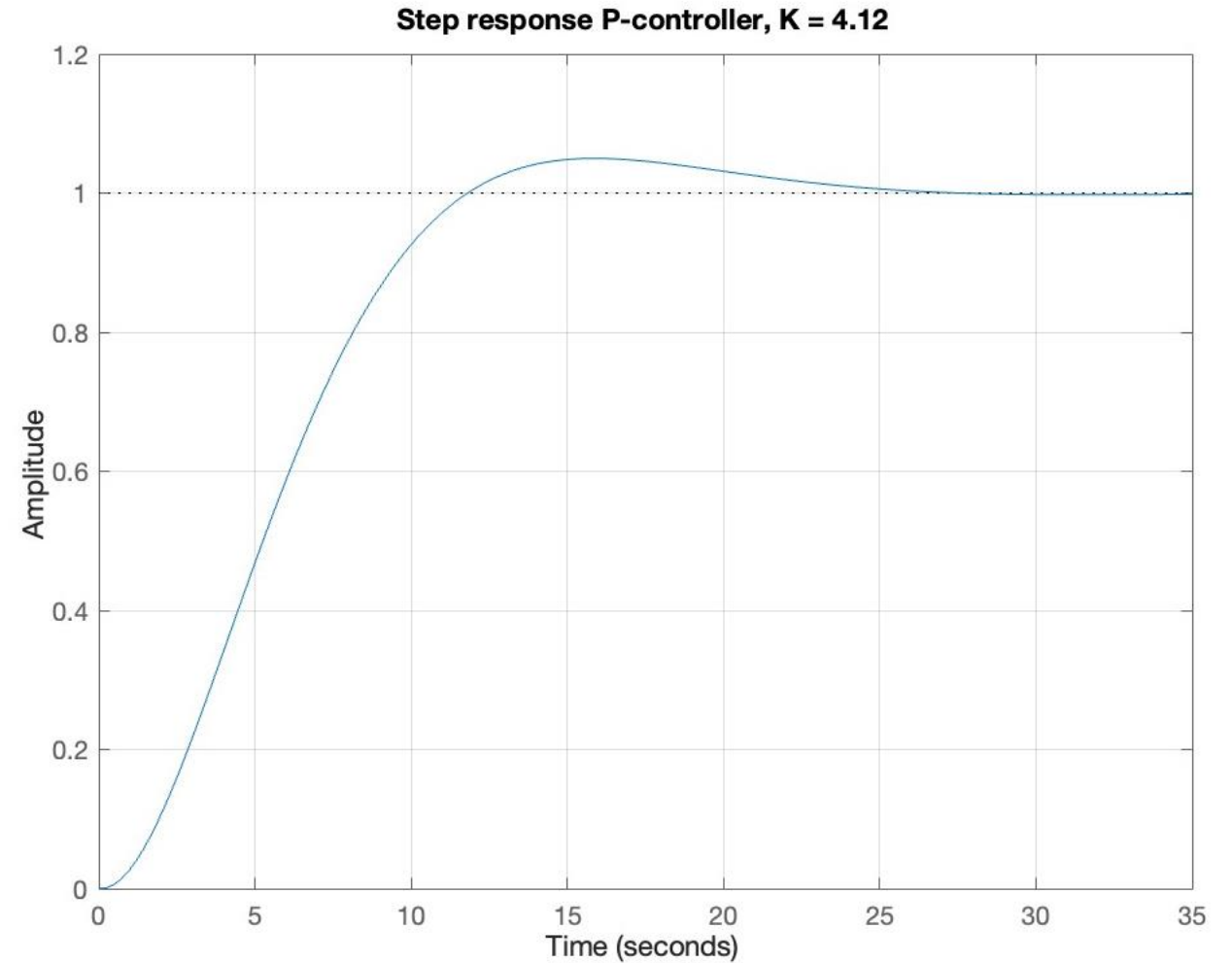
Assignment 2: Design of proportional controller

Requirements for the controller

- Overshoot $M < 5\%$
- Rise time as short as possible

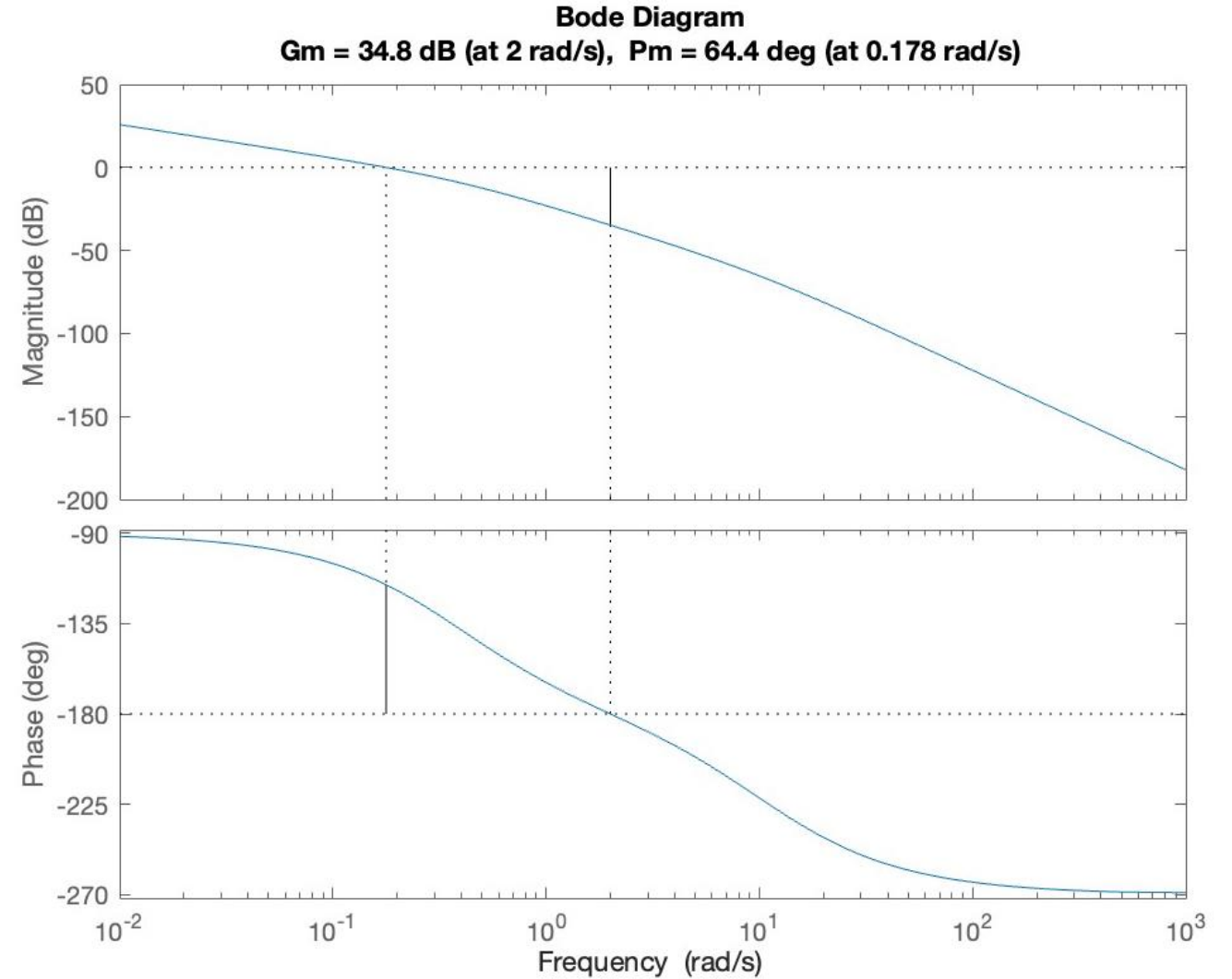
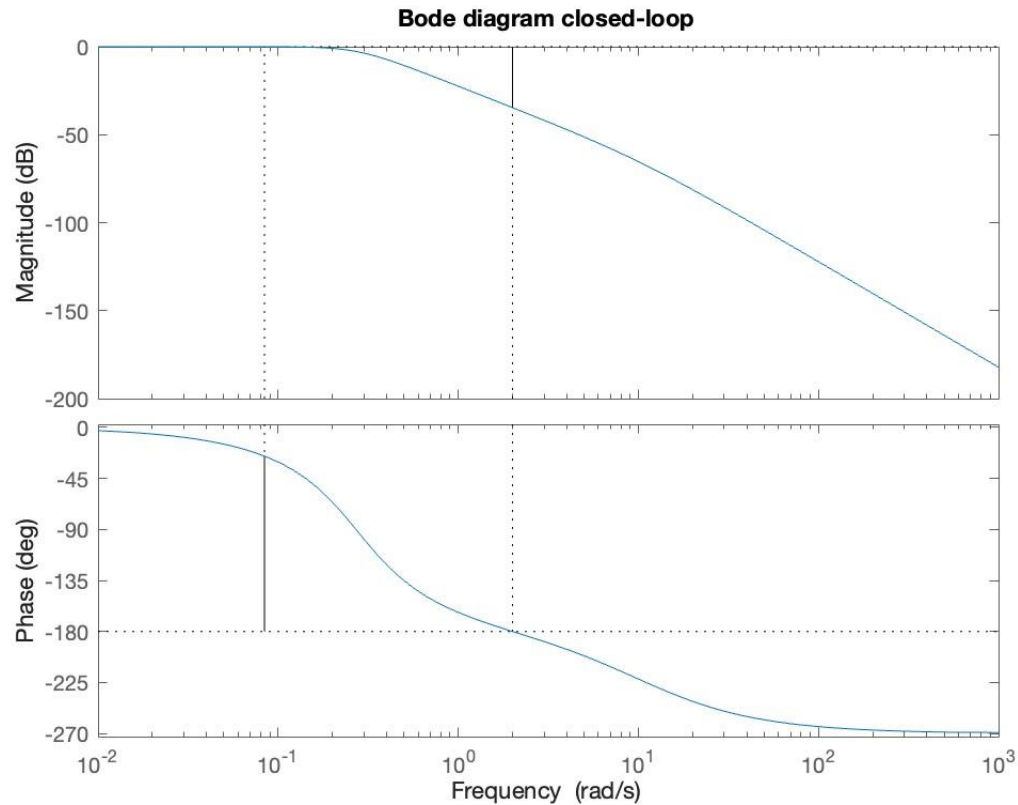
Best result accomplished with $K = 4.12$

- $M = 4.99\%$
- $T_r = 7.619s$



Assignment 3: Cross-over-frequency, phase margin, bandwidth

- Phase margin: $\varphi_m = 64.3786^\circ$
- Cross-over-frequency: $\omega_c = 0.1779 \frac{\text{rad}}{\text{s}}$
- Bandwidth: $\omega_b = 0.28$



Bode diagram for the open loop system

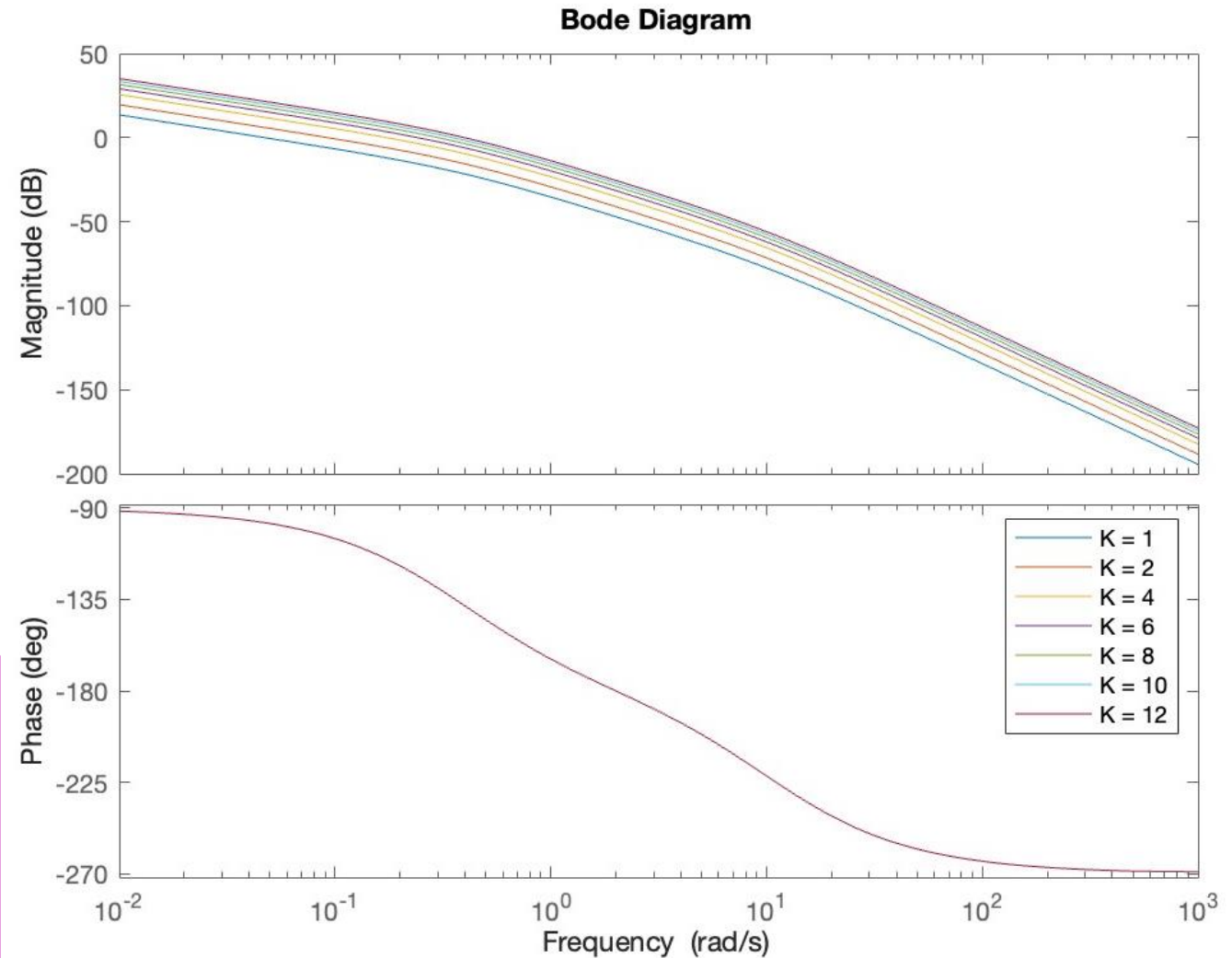
Assignment 4.1: What happens to the magnitude and phase plots when implementing a P-controller

K	φ_m	GM	ω_c	ω_p
1	82.81	225.3	0.0471	2.0
2	76.09	112.6	0.0924	2.0
4	64.93	56.32	0.1734	2.0
6	56.72	37.54	0.2418	2.0
8	50.58	28.16	0.3003	2.0
10	45.85	22.53	0.3517	2.0
12	42.08	18.77	0.3977	2.0

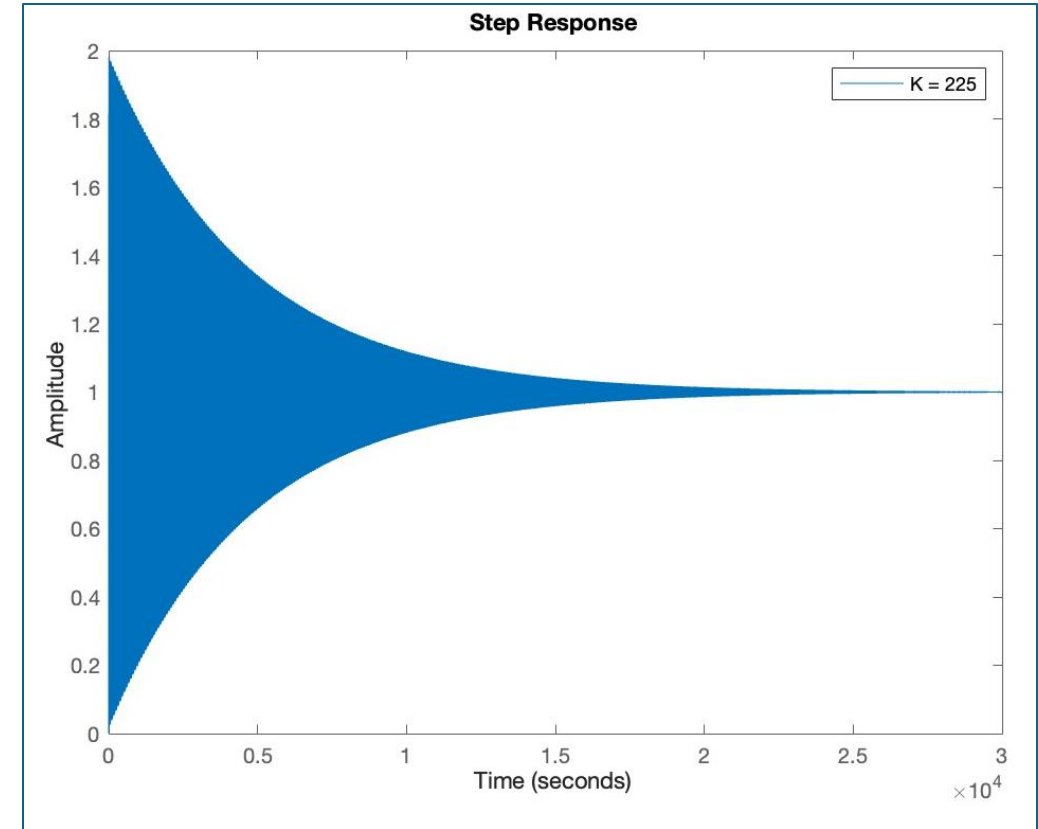
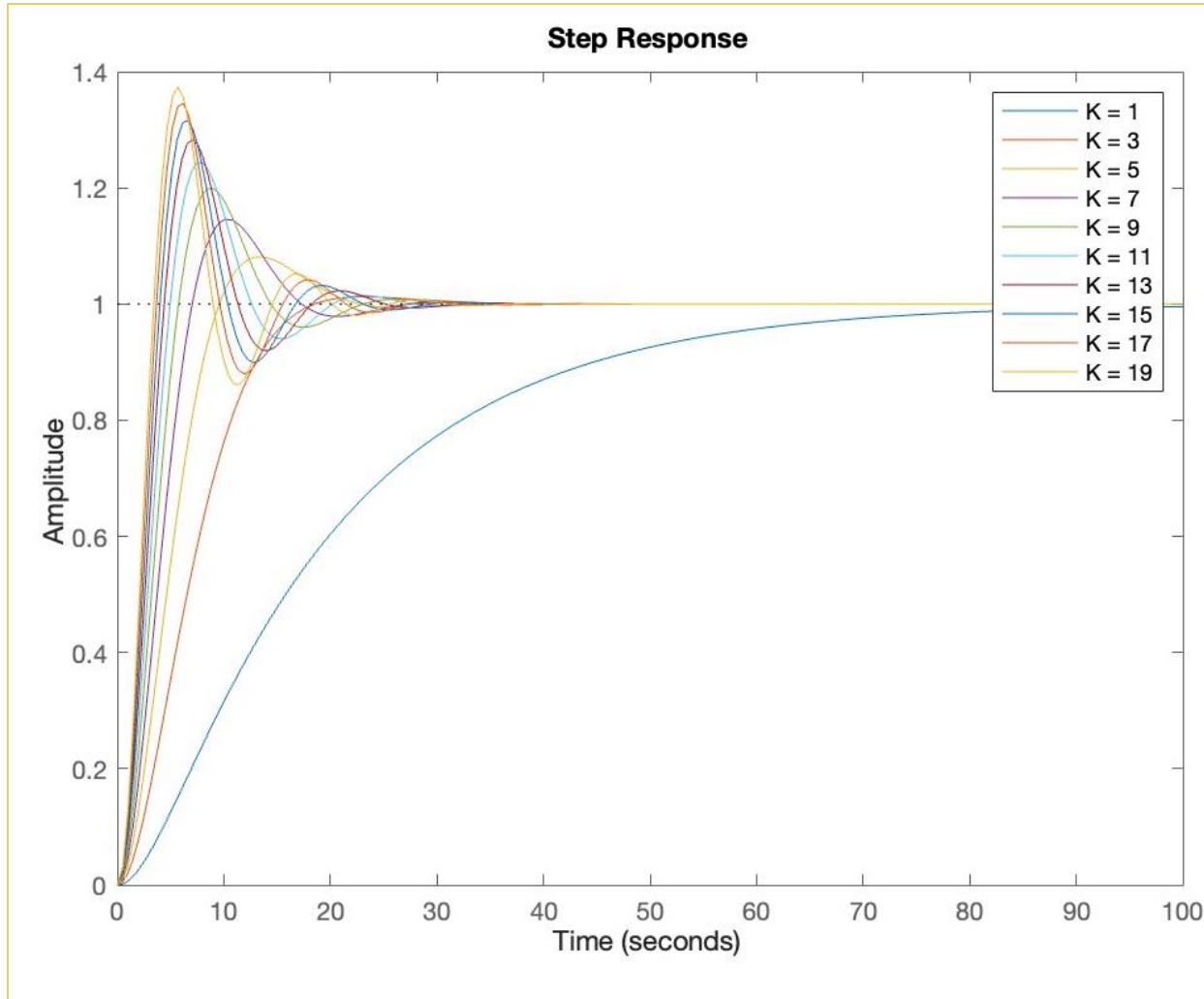
The magnitude response gets a constant gain added.

Higher cross-over frequency $\rightarrow$ faster system but lower phase margin $\rightarrow$ less stable / oscillation

➤ When implementing a P-controller it will increase the gain, but the phase plot will stay the same.



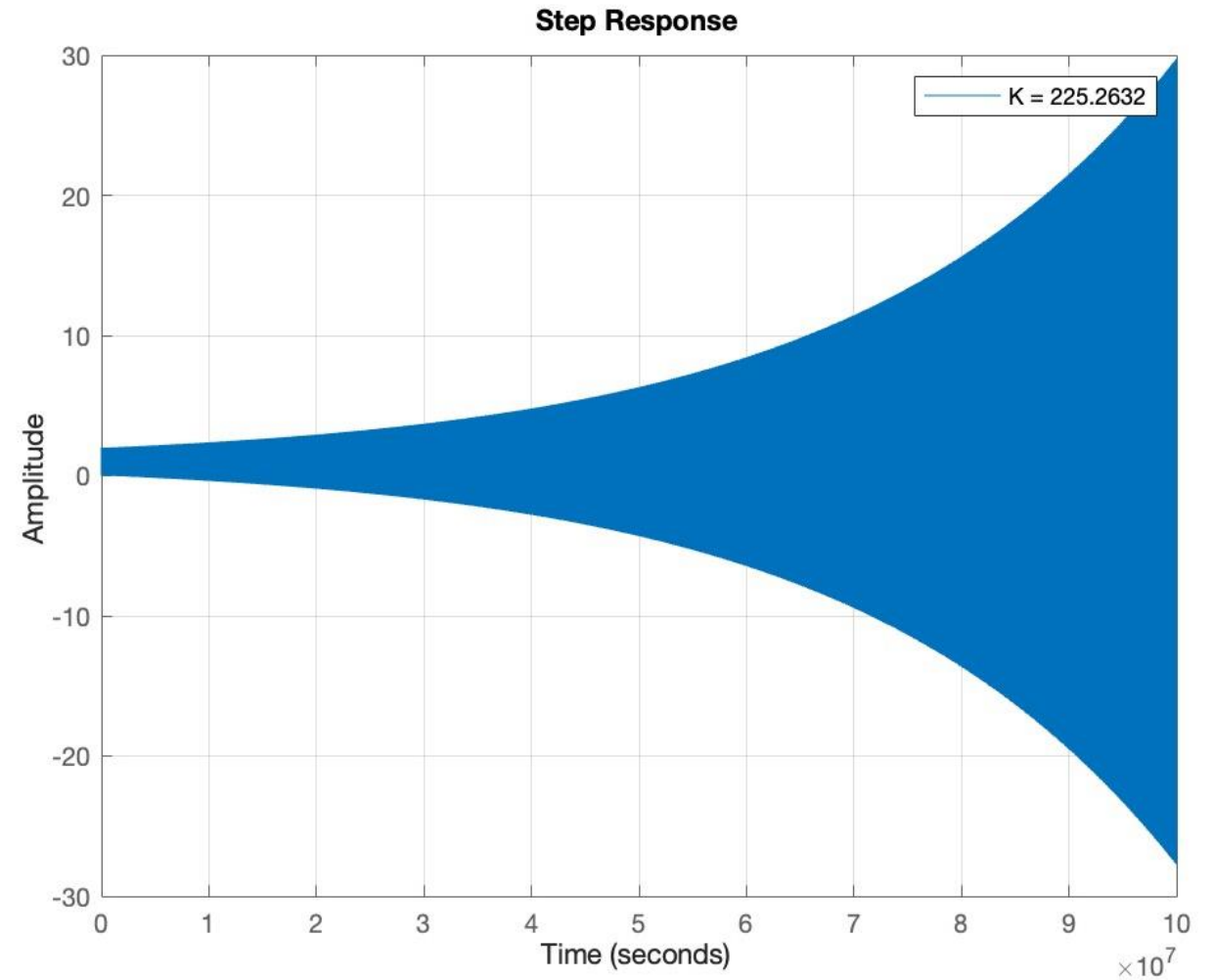
Assignment 4.2: What is limiting the speed of the system if ignoring the criteria of $M < 5\%$



When increasing K , the response is faster, but oscillations will increase and the system will at some K become unstable. That is when the gain margin is 0 dB/ phase margin = 0.

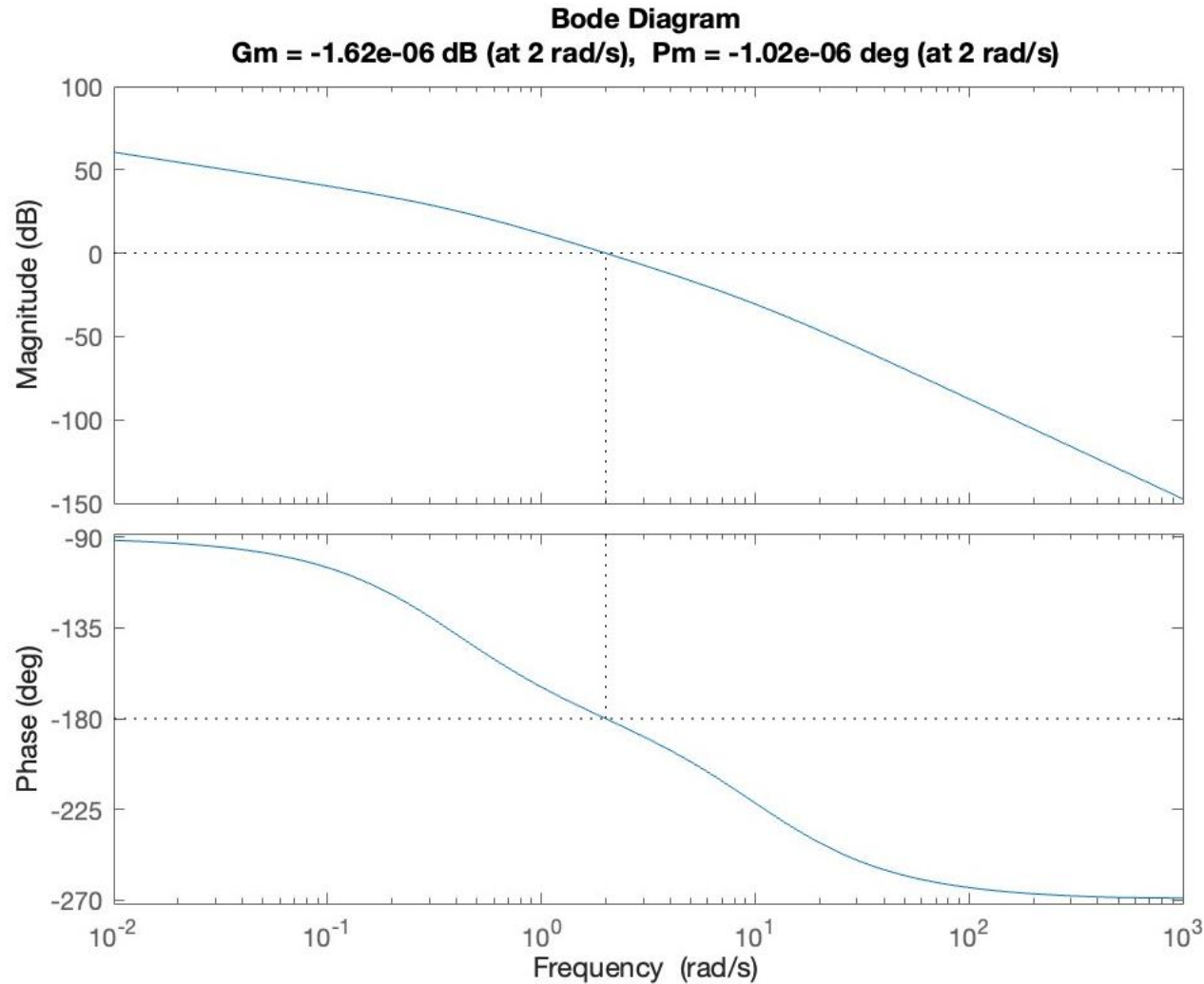
Assignment 4.3: What happens to the robot arm when the closed system loop becomes unstable?

When reaching an unstable condition, ($K > 225.2632$), the robot arm will oscillate uncontrollably, and the output signal will start to diverge.

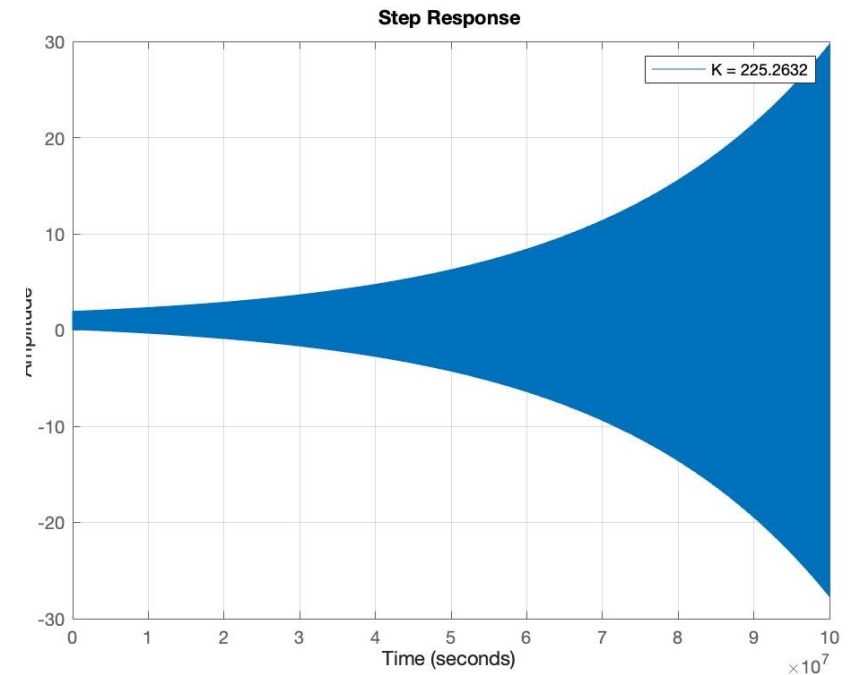


Assignment 4.4: At what gain K does the system become unstable? What happens to the phase margin?

The system becomes unstable when $K \geq 225.2632$

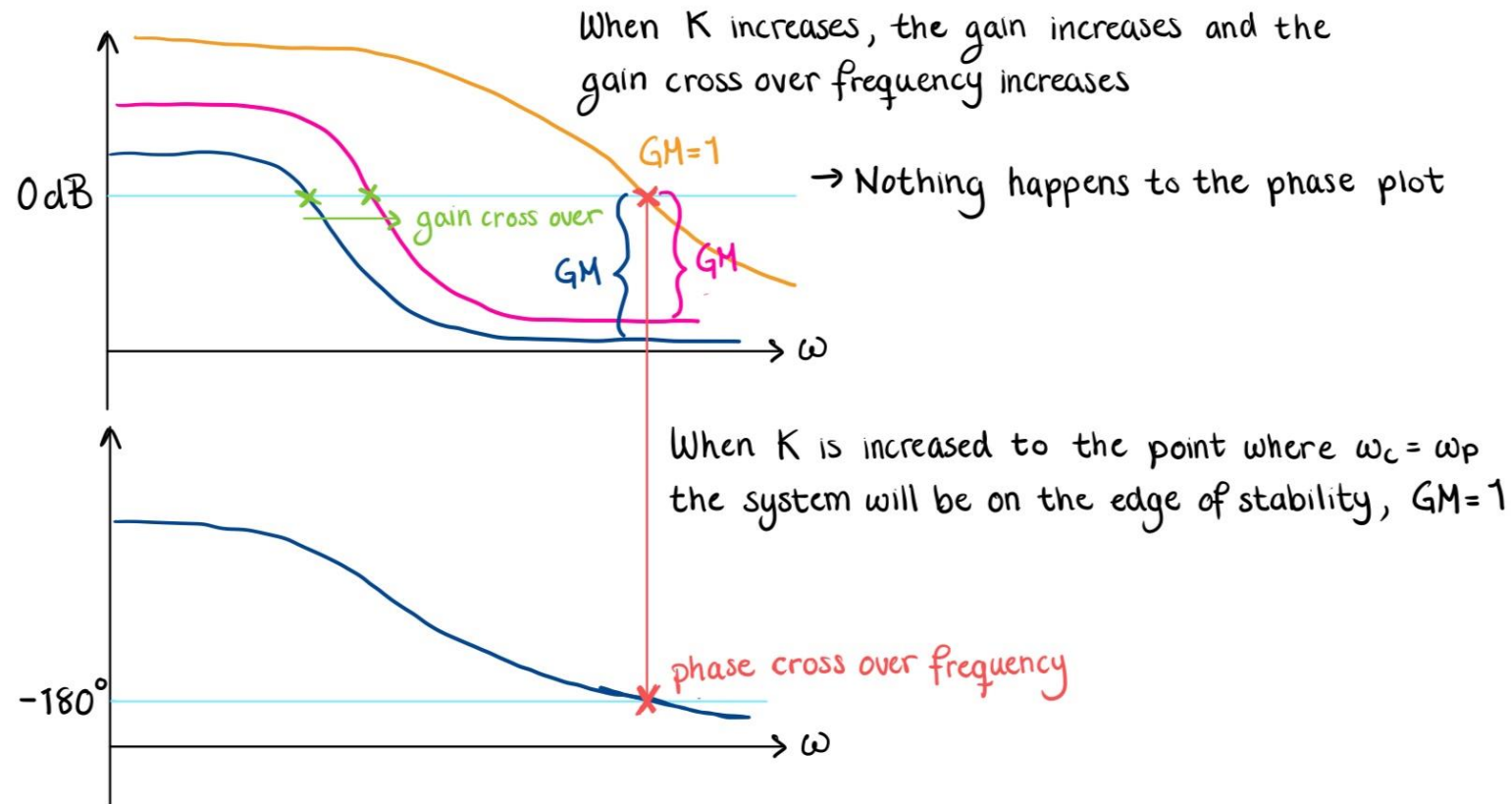


$$\varphi_m = -1.02 \cdot 10^{-6} \text{ }^\circ \approx 0^\circ$$

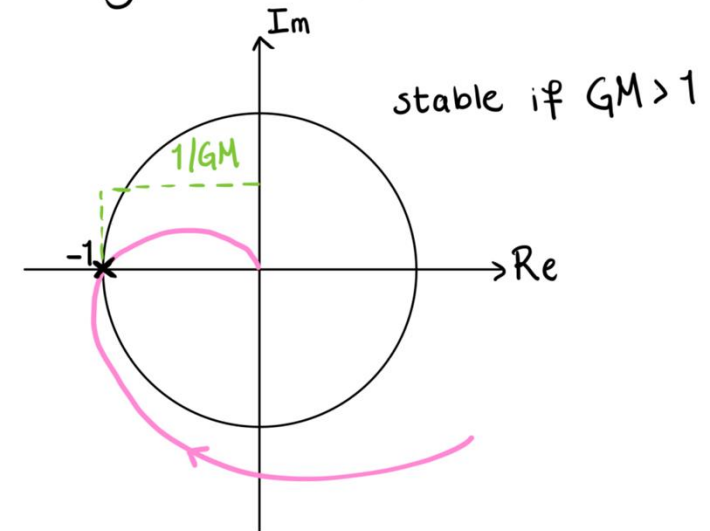


Assignment 4.5: What part of a Bode diagram/Nyquist curve indicates how much the gain can be increased before instability occurs?

You look at the gain margin. If $GM > 1$ the system is stable



open loop curve passes exactly through the critical point -1 if $\omega_p = \omega_c$



Assignment 5.1: Which frequency domain property of the open-loop system is most closely related to the rise time of the system? How should this be changed in the case with a lead-lag compensator to make it four times faster than the P-controller?

In the frequency domain, the bandwidth of the closed-loop system is directly proportional to the rise time of the system, for the open system the gain cross-over frequency is almost equal to the closed-loop system's bandwidth.

$$T_r \sim \frac{1}{\omega_b} \sim \frac{1}{\omega_c}$$

To make the system four times faster, we must increase the gain cross-over frequency/bandwidth by a factor of four.

Assignment 5.2: What frequency domain property of the open-loop system should be kept approximately the same to make sure the overshoot is under 5%?

The resonance peak (M_p) is tied to the overshoot (M_y) of the system. To keep the resonance peak not get too high we have to keep the phase margin of the open-loop system approximately the same.

$$M_y \sim M_p \sim \frac{1}{\varphi_m}$$

Assignment 6: Determine the transfer function $F(s)$ of a lead-lag compensator that satisfies the new specifications.

LEAD

cross-over frequency

Want 4 times faster,

$$\omega_{c,new} = 4\omega_c = 0.7116 \text{ rad/s}$$

phase margin:

With $\omega_{c,new} = 4\omega_c$ we get

$$\varphi_m = 180^\circ - 155.35^\circ = 24.65^\circ$$

We need $M < 5\%$, $M \sim \frac{1}{\varphi_m}$, want to keep approximately the same phase margin (64.38°)

$$\Rightarrow \varphi_m = 24.65^\circ + 39.73^\circ + 5.7^\circ = 70.08^\circ$$

lag-part decrease the phase $\sim 5.7^\circ$

β : Look at p. 106 in course literature,

$$\alpha \quad \varphi_{\max} = 39.73^\circ + 5.7^\circ = 45.43^\circ \text{ represents a } \beta \approx 0.16$$

$$\tau_D: \omega_c = \frac{1}{\tau_D \sqrt{\beta}} \Leftrightarrow \tau_D = \frac{1}{\omega_c \sqrt{\beta}}$$

$$K: \text{ Want } K \text{ so that } |G(i\omega_{c,lead})| = \frac{K}{\sqrt{\beta}}$$

$$\Rightarrow K = \frac{\sqrt{\beta}}{|G(i\omega_{c,lead})|}$$

LAG:

specify δ

We have an integrated step

$$\Rightarrow e_1 = \lim_{s \rightarrow 0} \frac{1}{sF(s)G(s)}$$

$$\lim_{s \rightarrow 0} sF_{lead} G(s) = K \frac{\tau_D s + 1}{\beta \tau_D s + 1} \frac{1.9s}{10s^3 + 107s^2 + 40s} =$$

$$= K \frac{\overset{0}{\tau_D s + 1}}{\underset{0}{\beta \tau_D s + 1}} \frac{1.9}{\underset{0}{10s^2 + 107s + 40}} = K \frac{1.9}{40}$$

$$\Rightarrow e_1 = \frac{40}{1.9K} < 0.05 \Rightarrow \text{decrease error: } \frac{e_1}{0.05}$$

$$\Rightarrow \gamma = \frac{1}{\frac{e_1}{0.05}} = \frac{0.05}{e_1}$$

τ_I : Try τ_I to get overshoot $< 5\%$

$$\text{first tried with } \tau_I = \frac{10}{\omega_{c,lead}} \Rightarrow M > 5\%$$

$$\text{increase the integrating factor: } \tau_I = \frac{20}{\omega_c}$$

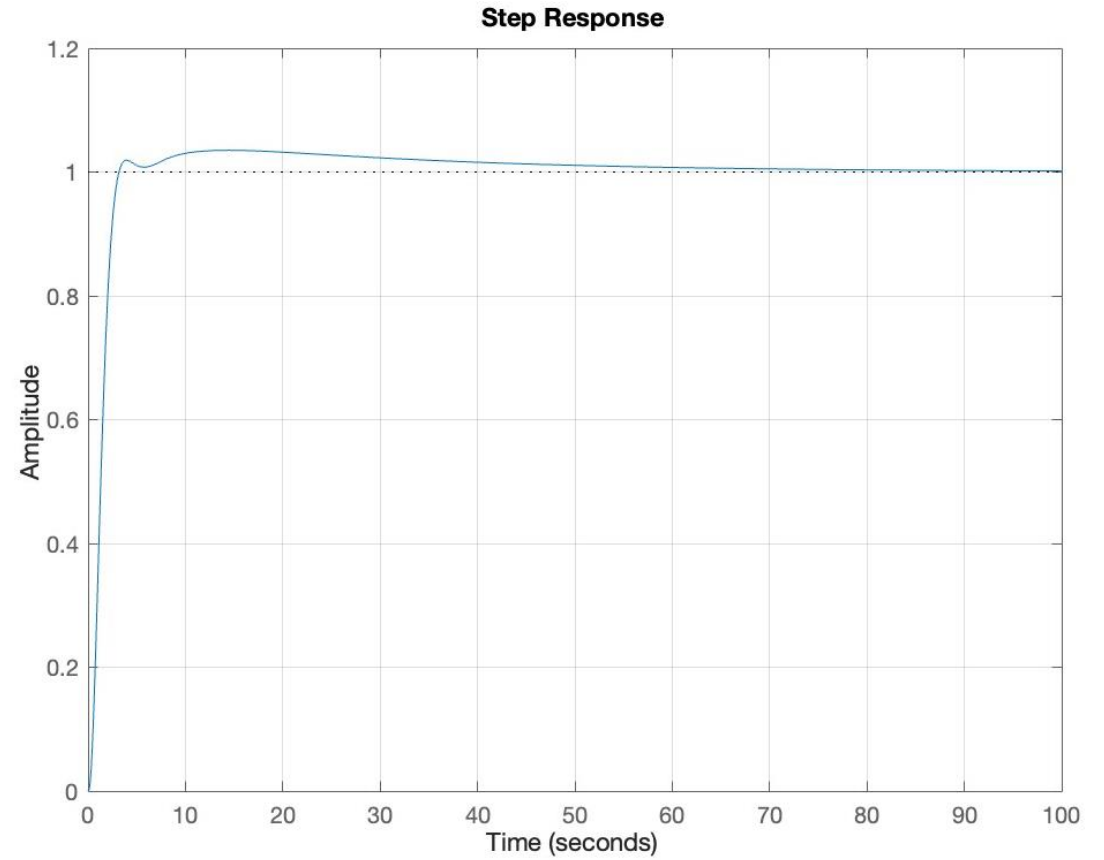
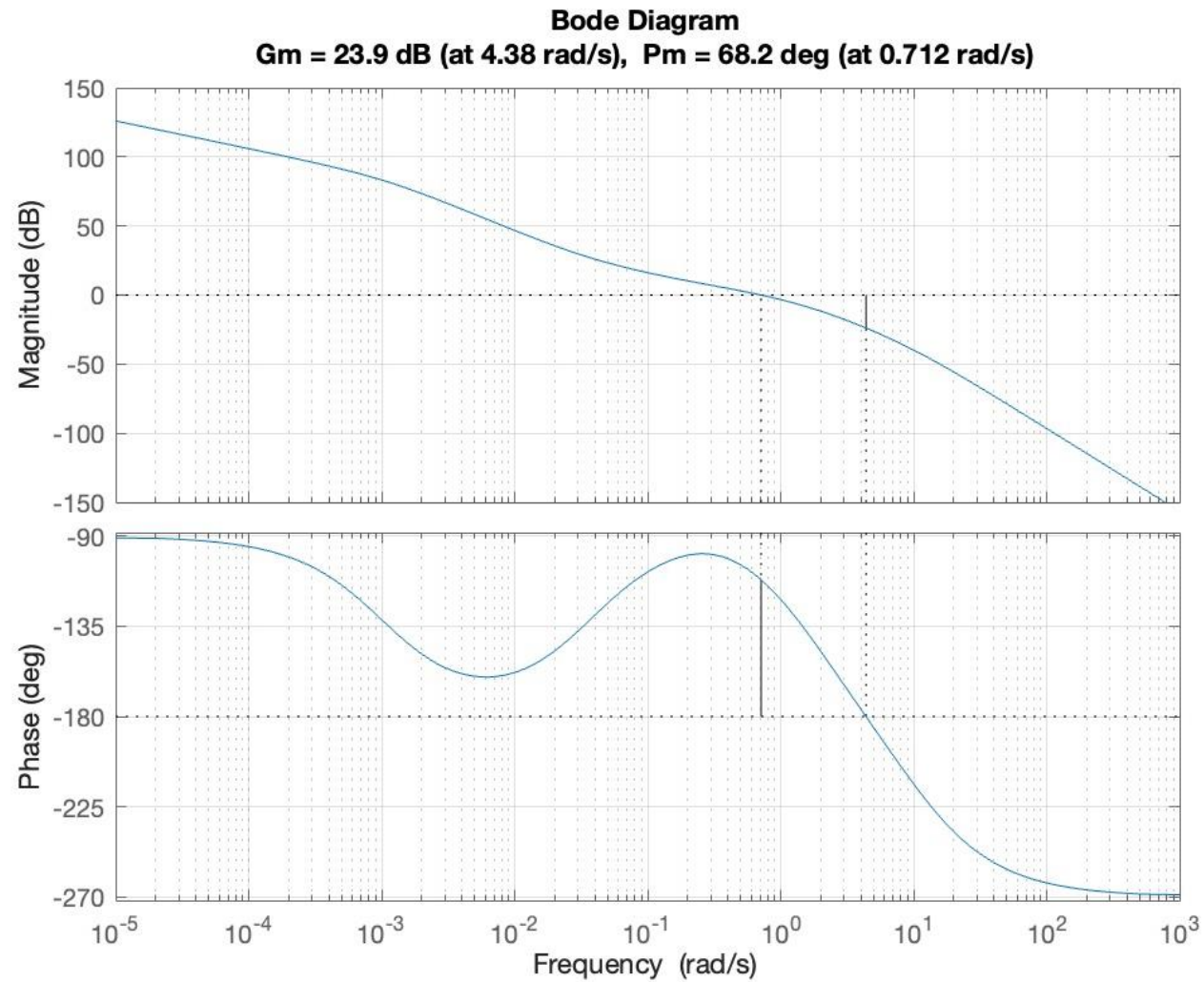
checking that $u(t) < u_{\max} \forall t$

$$\text{transfer function } r \rightarrow u: G_{ru} = \frac{1}{1+FG}$$

$F_{lead_lag} =$

$$\frac{1239 s^2 + 396.8 s + 12.55}{15.8 s^2 + 28.12 s + 0.02981}$$

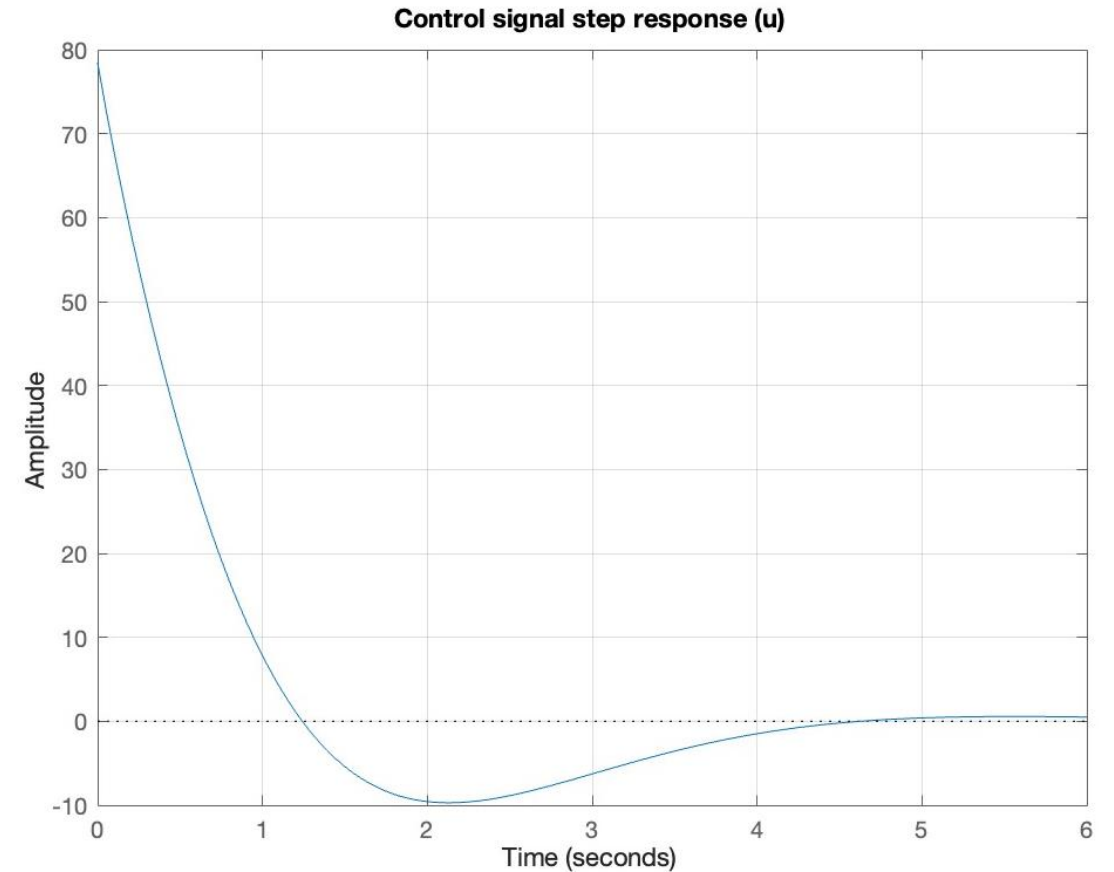
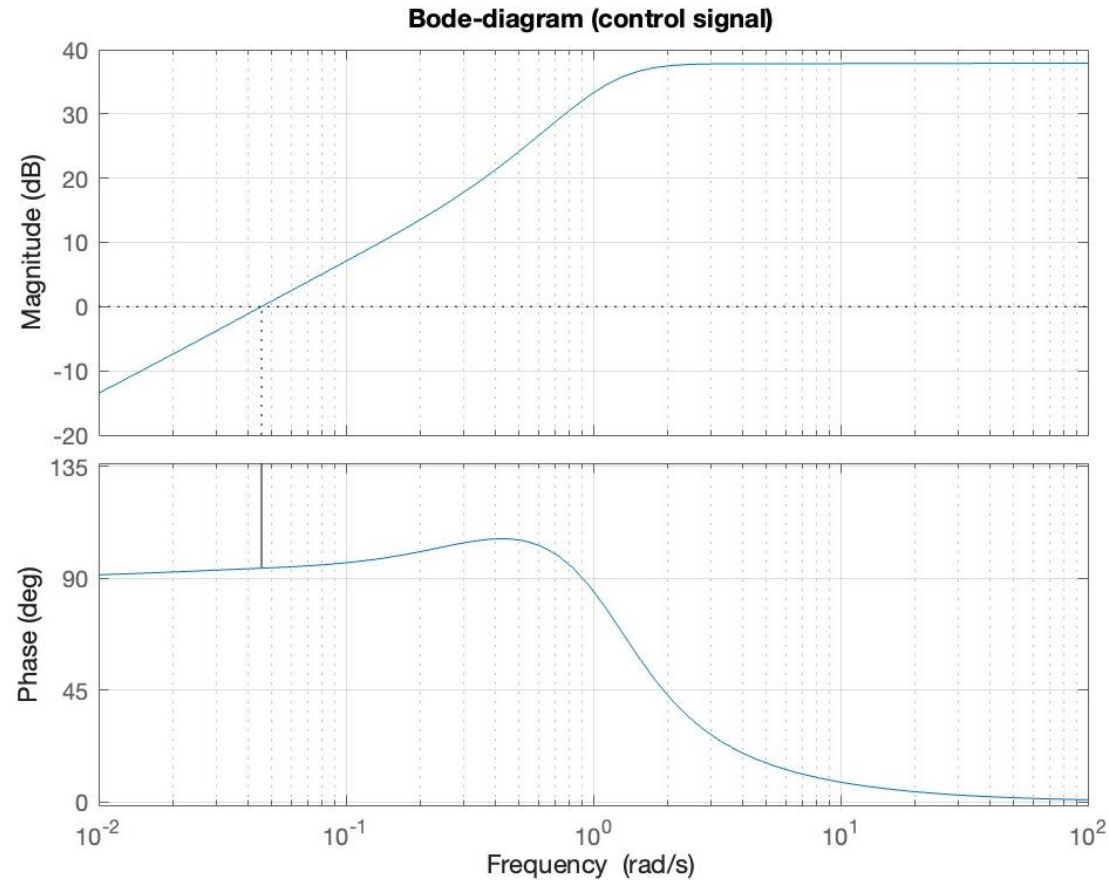
Assignment 6: Determine the transfer function $F(s)$ of a lead-lag compensator that satisfies the new specifications.



$$M_y = 3.51\%$$

$$T_r = 1.892 \text{ s}$$

Assignment 6: Determine the transfer function $F(s)$ of a lead-lag compensator that satisfies the new specifications.



The high gain (~ 40 dB) at high frequencies is a necessary trade-off to meet the speed requirement. This amplifies the noise at high frequencies (100x). It is sufficient in simulation but in real life it can lead to physical implications

Assignment 7: Why is the control error for a ramp input not reduced to zero?

- The control error for a ramp input is not reduced to zero because the open-loop system $G_o = F_{lead-lag} \cdot G$ only has one integrator.

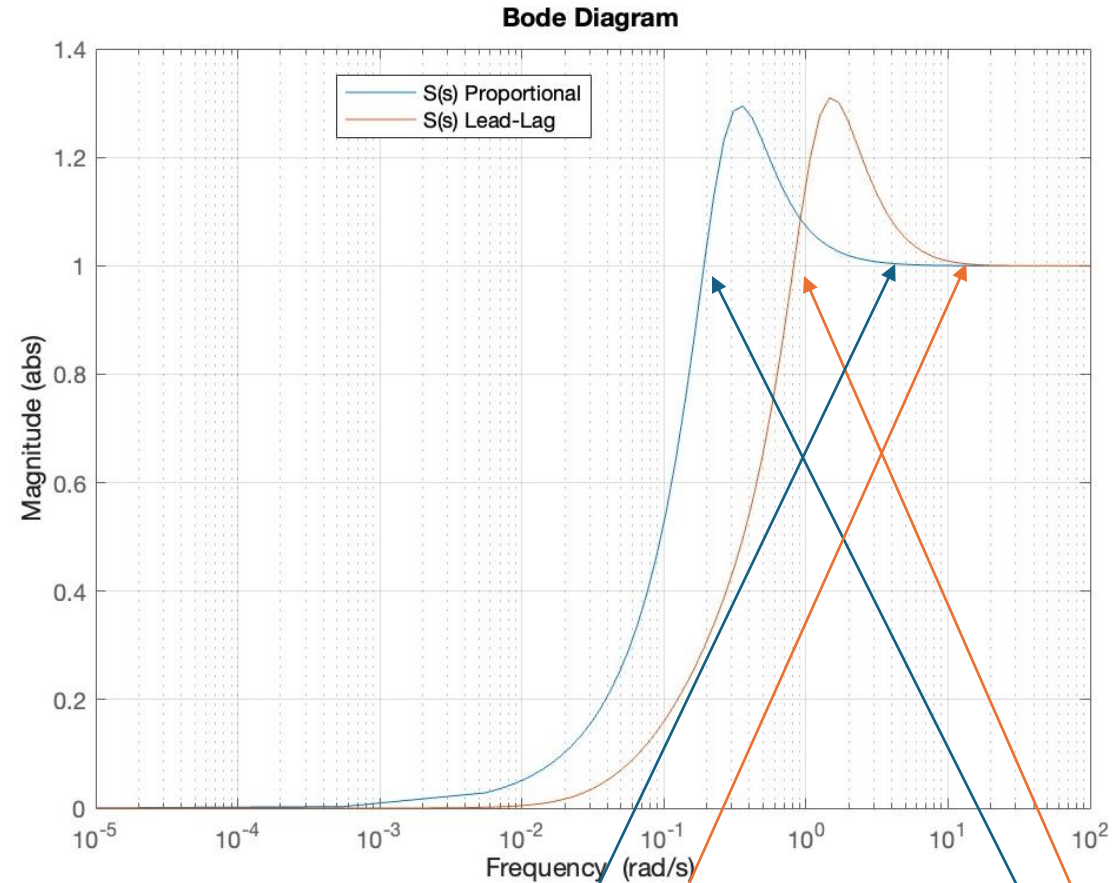
A ramp input requires a system with two integrators $\left(\frac{1}{s^2}\right)$ to have zero control error. The system G has one integrator $\left(\frac{1}{s}\right)$, and the lead-lag compensator has no integrator (since $\gamma \neq 0$).

Assignment 8: Calculate and plot the Bode diagram of the sensitivity functions $S(s)$ for both controllers. For which frequencies do the controllers attenuate the effect of disturbances on the output? For which frequencies are the effect of disturbances amplified?

$$S(s) = \frac{1}{1+F(s)G(s)} = \frac{1}{1+G_0}$$

Attenuation: The controllers attenuate disturbances at low frequencies. This corresponds to the frequency range where the magnitude of the sensitivity function, $|S(s)| < 1$.

Amplification: Disturbances are amplified at intermediate frequencies. This corresponds to the "peak" in the plot where the magnitude of $|S(s)| > 1$.



Amplification:

$$S(s)_{prop}: 0.192 < \omega < 3.85$$

$$S(s)_{lead-lag}: 0.82 < \omega < 12.3$$

Attenuation:

$$S(s)_{prop}: 0.192 > \omega$$

$$S(s)_{lead-lag}: 0.82 > \omega$$

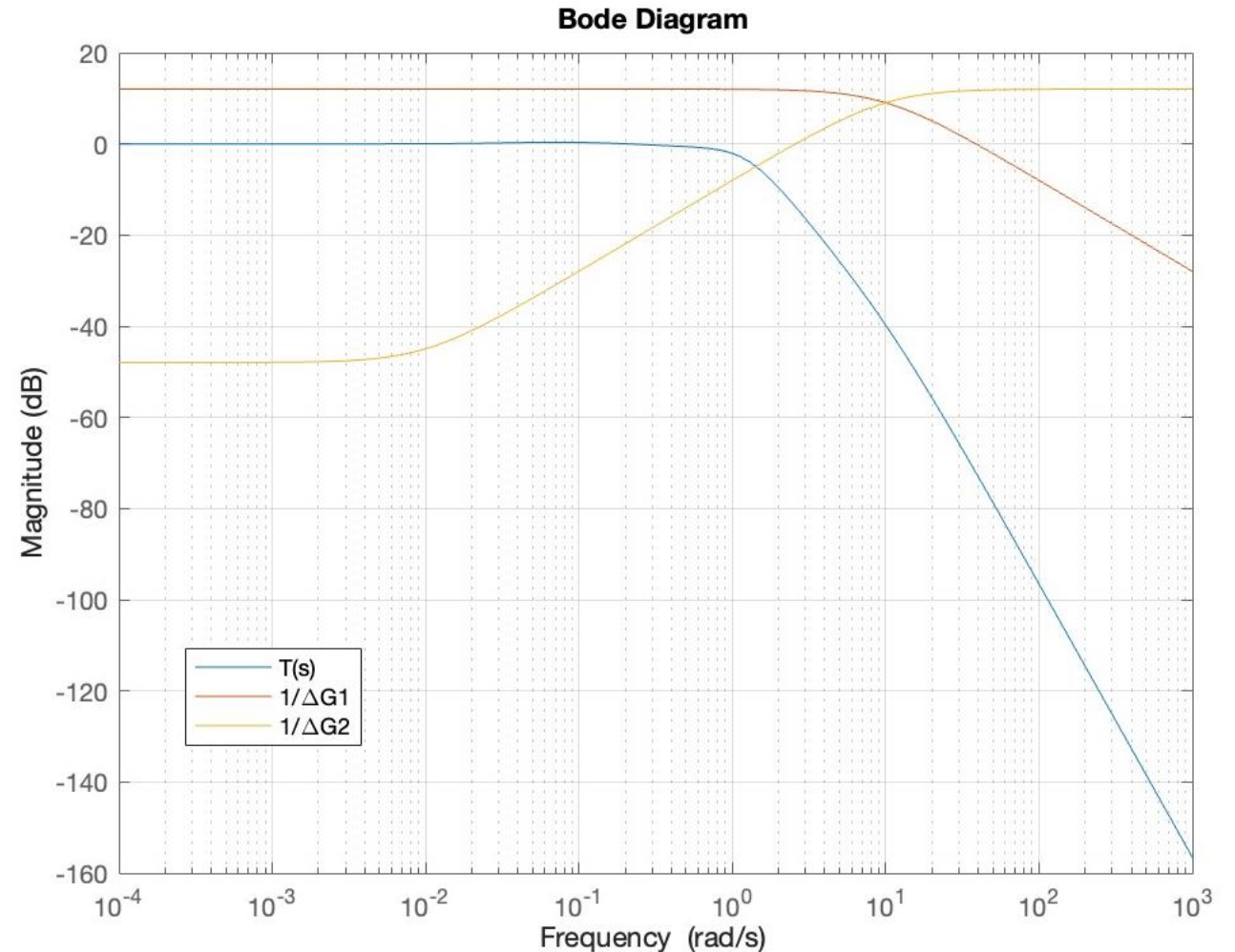
Assignment 9: Will the robustness criterion guarantee robust stability for both model error descriptions?

Robustness criterion: $|T(s)| < \frac{1}{|\Delta G(s)|} \quad \forall \omega$

$$\Delta G_1(s) = \frac{s + 10}{40} \quad \text{and} \quad \Delta G_2(s) = \frac{s + 10}{4(s + 0.01)}$$

We can see in the Bode diagram that the criterion is fulfilled for $\Delta G_1(s)$.

It is not fulfilled for $\Delta G_2(s)$ – so stability is not guaranteed.



Assignment 10: Find the state space description of the system

$$x_1 = \theta_L, \quad x_2 = \dot{\theta}_m, \quad x_3 = I_a$$

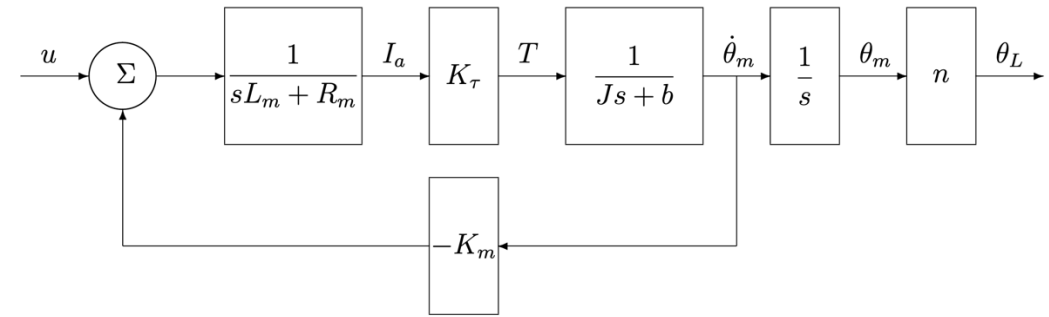
Want to find the matrices that fulfill
 $\dot{x}(t) = A x(t) + B u(t), \quad y(t) = C x(t) + D$

From block diagram

$$\theta_L = n \frac{1}{s} \dot{\theta}_m \Leftrightarrow s \theta_L = n \dot{\theta}_m$$

$$\dot{\theta}_m = \frac{1}{J s + b} K_\tau I_a \Leftrightarrow s \dot{\theta}_m = -\frac{1}{J} b \dot{\theta}_m + \frac{1}{J} K_\tau I_a$$

$$I_a = \frac{1}{s L_m + R_m} (u - K_m \dot{\theta}_m) \Leftrightarrow s I_a = -\frac{1}{L_m} R_m I_a + \frac{u}{L_m} - \frac{K_m}{L_m} \dot{\theta}_m$$



Inverse Laplace transforms:

$$\dot{x}_1 = n x_2$$

$$\dot{x}_2 = -\frac{b}{J} x_2 + \frac{K_\tau}{J} x_3$$

$$\dot{x}_3 = -\frac{R_m}{L_m} x_3 + \frac{1}{L_m} u - \frac{K_m}{L_m} \dot{\theta}_m$$

$$\Rightarrow \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \overbrace{\begin{bmatrix} 0 & n & 0 \\ 0 & -b/J & K_\tau/J \\ 0 & -K_m/L_m & -R_m/L_m \end{bmatrix}}^A \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \underbrace{\begin{bmatrix} 0 \\ 0 \\ 1/L_m \end{bmatrix}}_B u$$

We have that $y = \theta_L = x_1$

$$y = \overbrace{[1 \ 0 \ 0]}^C \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \underbrace{0}_{\leftarrow D}$$

Assignment 11: Is the system controllable and observable? Motivate

Controllability Matrix:

$$S = [B \quad AB \quad A^2B]$$

Observability Matrix:

$$O = \begin{bmatrix} C \\ CA \\ CA^2 \end{bmatrix}$$

Controllability means the input can steer the system to *any* desired state, while observability means the internal states can be fully reconstructed just by measuring the output.

We check the rank to verify that the vectors in the controllability and observability matrices are linearly independent. Since the system has 3 states, a rank of 3 guarantees that these vectors span the entire state space, meaning no part of the system's dynamics is outside our control or observation.

The system is controllable and observable because the matrices has full rank (3), which equals the number of system states (x_1, x_2, x_3).

Assignment 12: Use state feedback and pole-placement to find a control law which gives a closed loop system that satisfies the requirements.

Pole placement : $s = -\zeta\omega_n \pm i\omega_n\sqrt{1 - \zeta^2}$

- 1) Fast as possible $\rightarrow$ controlled by the distance from origin, further away = faster $\rightarrow$ larger $u(t)$
- 2) $M < 5\%$ $\rightarrow$ controlled by damping $\zeta = \cos\theta$, want $\zeta \approx 0.7$ ($\theta = 45^\circ$)
- 3) $|u(t)| < u_{\max} \rightarrow$ Poles can not be too far
- 4) static gain = 1 $\rightarrow$ controlled by l_0

The third pole (s_1 in MATLAB): step response $\Rightarrow y(t) = 1 - k_1 e^{-s_2 t} - k_2 e^{-s_1 t}$
If it is too close to the dominant poles (s_2, s_3) it will "disturb" the step response
 $\rightarrow$ Put it further away

Assignment 12: Use state feedback and pole-placement to find a control law which gives a closed loop system that satisfies the requirements.

Let $u = -Lx + l_0 r$, $L = [l_1 \ l_2 \ l_3]$, $l_0 \in \mathbb{R}$

$$\dot{x} = Ax + B(-Lx + l_0 r) = (A - BL)x + Bl_0 r$$

$$P(s) = \det(sI - (A - BL)) = \det \begin{bmatrix} s & -n & 0 \\ 0 & s + b/J & -K_r/J \\ l_1/L_m & \frac{K_m + l_2}{L_m} & s + \frac{R_m + l_3}{L_m} \end{bmatrix} = (s - s_1)(s - s_2)(s - s_3)$$

$$= s \det \begin{bmatrix} s + b/J & -K_r/J \\ \frac{K_m + l_2}{L_m} & s + \frac{R_m + l_3}{L_m} \end{bmatrix} - 0 + \frac{l_1}{L_m} \det \begin{bmatrix} -n & 0 \\ s + b/J & -K_r/J \end{bmatrix} =$$

$$= s \left(\left(s + \frac{b}{J}\right) \left(s + \frac{R_m + l_3}{L_m}\right) + \frac{K_r}{J} \cdot \frac{K_m + l_2}{L_m} \right) + \frac{l_1}{L_m} \frac{n K_r}{J}$$

$$= s^3 + s^2 \left(\frac{R_m + l_3}{L_m} + \frac{b}{J} \right) + s \left(\frac{b(R_m + l_3)}{J L_m} + \frac{K_r(K_m + l_2)}{J L_m} \right) + \frac{l_1}{L_m} \frac{n K_r}{J}$$

$\Rightarrow$ This is done in MATLAB with $L = \text{place}(A, B, \text{poles}) = [l_1, l_2, l_3]$

Determining l_0 : static gain = 0 $\Rightarrow y = r$ when $t \rightarrow \infty$

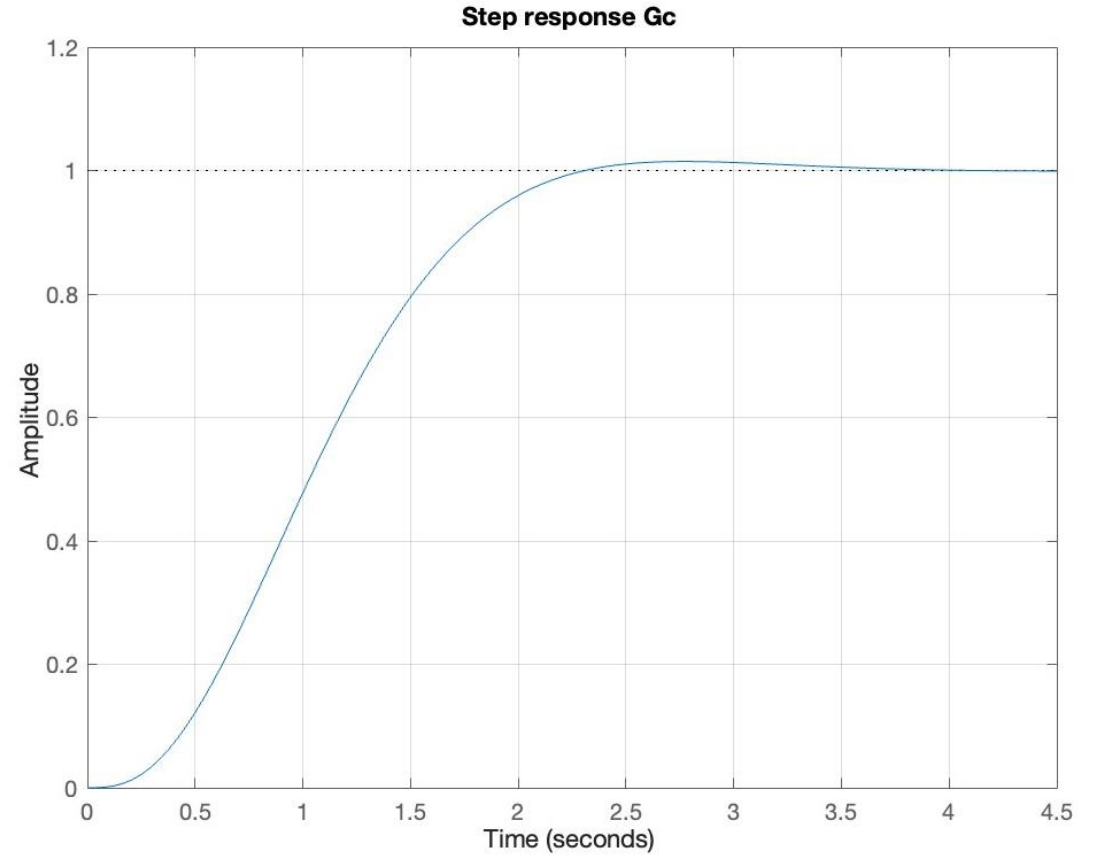
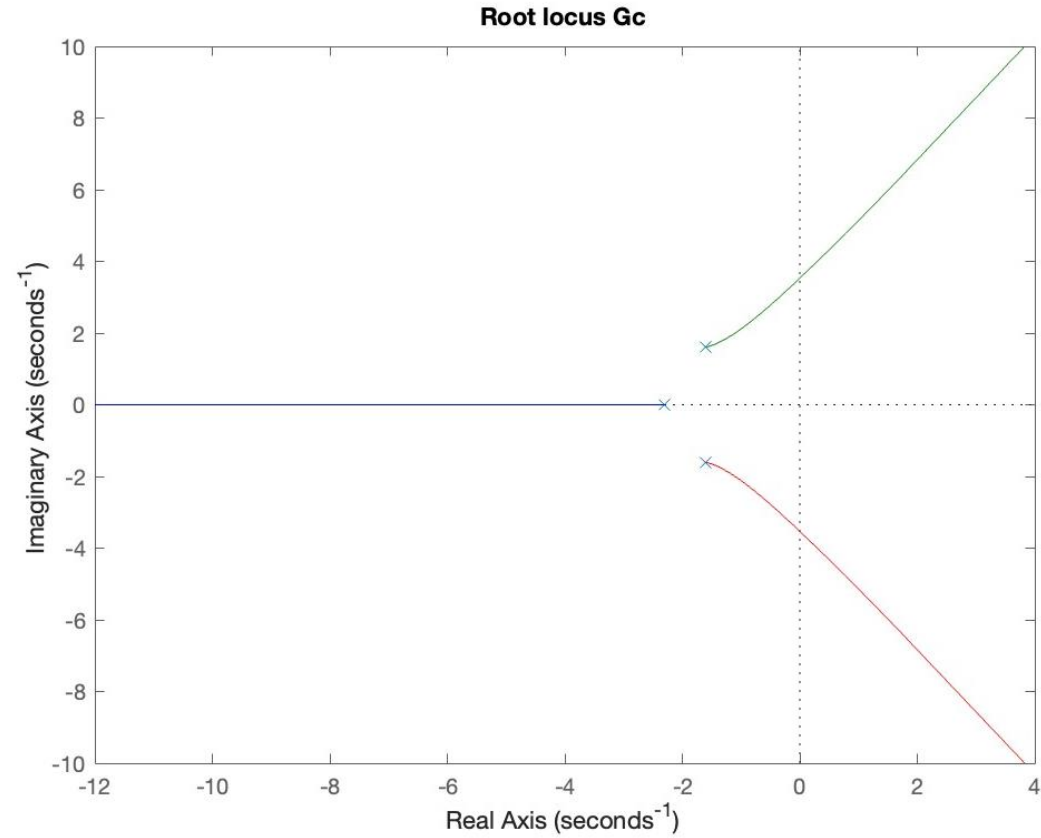
at equilibrium: $\dot{x} = (A - BL)x + Bl_0 r = 0$

$$\Rightarrow -(A - BL)x = Bl_0 r \Rightarrow x = -(A - BL)^{-1} Bl_0 r$$

$$y = Cx \Rightarrow y = C[-(A - BL)^{-1} Bl_0 r]$$

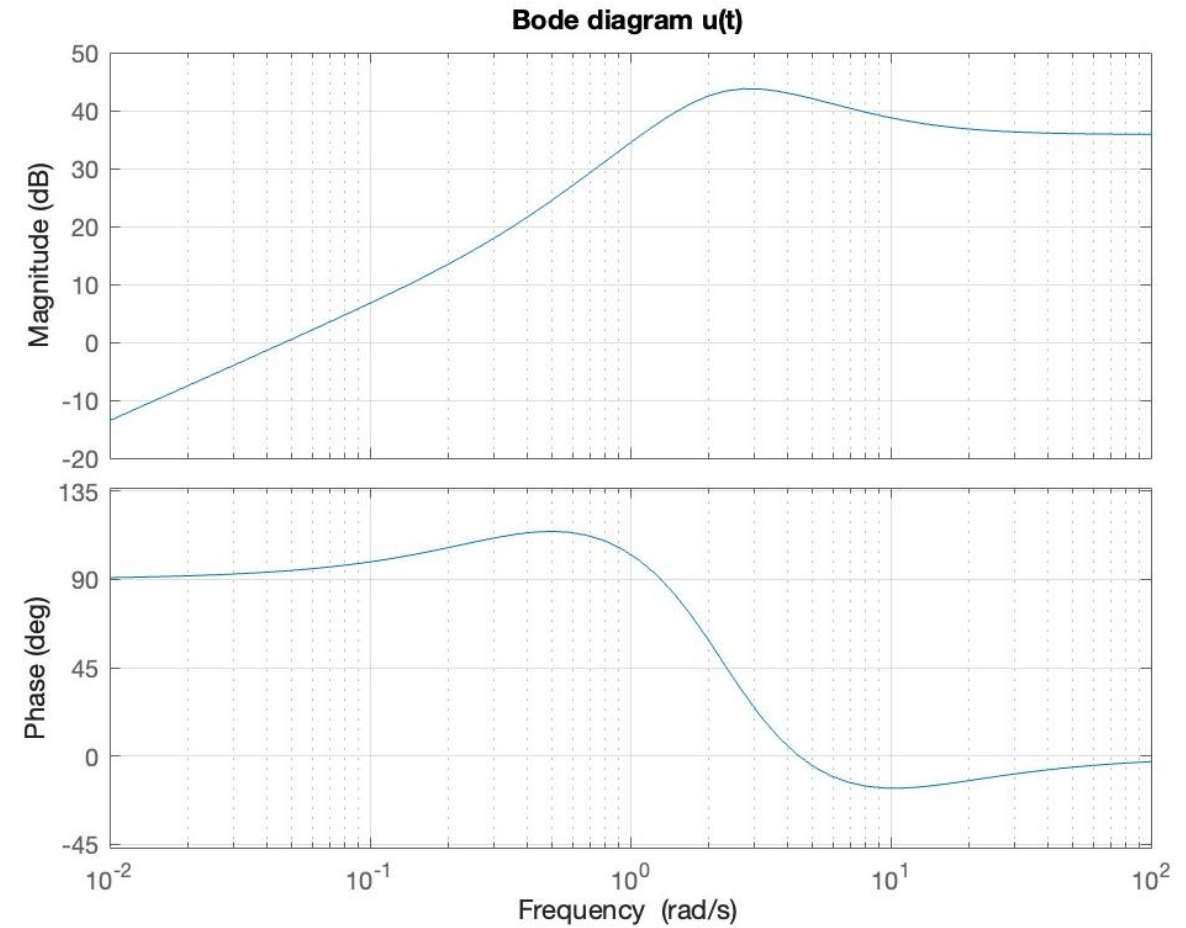
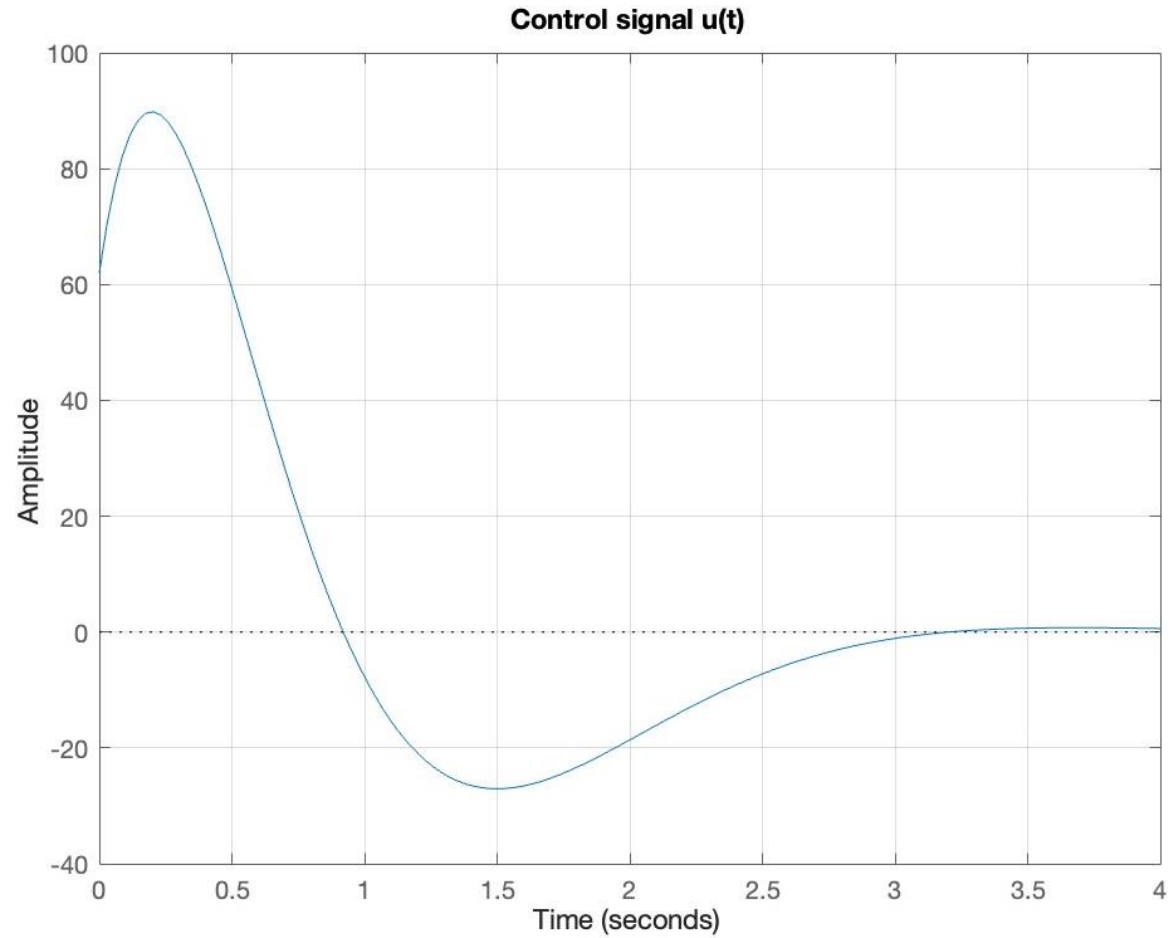
$$\text{for } y = r \Rightarrow -C(A - BL)^{-1} B \cdot l_0 = 1 \Rightarrow l_0 = \frac{1}{-C(A - BL)^{-1} B}$$

Assignment 12: Use state feedback and pole-placement to find a control law which gives a closed loop system that satisfies the requirements.



$$M = 1.5 \% \quad T_r = 1.304 \text{ s}$$

Assignment 12: Use state feedback and pole-placement to find a control law which gives a closed loop system that satisfies the requirements.



Assignment 13: What is the effect of having a static gain of 1 in terms of reference tracking ($e=r-y$)?

Having a static gain of 1 ensures that the steady-state error for a step input is zero.

The static gain ratio between the output y and the reference r at steady state ($t \rightarrow \infty$). If this ratio is 1, then $y(\infty) = r(\infty)$. Consequently, the error $e(\infty) = r(\infty) - y(\infty) = 0$.

This is achieved by the constant l_0

Assignment 13: Motivate the choice of poles/feedback constants of the system.

We chose two complex conjugate poles as our dominant pair. This allows us to control the overshoot (by adjusting the ratio between the real and imaginary parts, i.e., the damping) and the speed (by adjusting the distance from the origin). We chose these poles to lie on the 45° , corresponding to a good damping ratio $\zeta \approx 0.7$.

We tuned the poles by moving them as far to the left as possible to maximize speed but stopped when the control signal approached its maximum.

We placed the third pole on the negative real axis, further away from the origin than the dominant poles (more negative real part). This ensures that the third pole's dynamics decay faster, preventing it from 'disturbing' the dominant poles and the designed overshoot.

Assignment 13: Which rise time is achieved and how does this compare to the lead-lag design in assignment 6?

Lead-lag compensator: $T_r = 1.892 \text{ s}$

State feedback controller: $T_r = 1.304 \text{ s}$

State feedback has access to more information (all internal states: position, velocity, current) compared to the lead-lag compensator which only "sees" the output angle. This allows the state feedback controller to push the system closer to its physical limits ($u_{\max}$) more aggressively while maintaining stability, whereas the lead-lag design is often limited by the trade-off between phase margin (overshoot) and bandwidth (rise time).

Assignment 13: What is the biggest difference between a lead-lag compensator and a state feedback controller in terms of physical implementation?

The biggest difference in physical implementation is that a lead-lag compensator operates only on the system's input and output signals, whereas a state feedback controller needs access to all internal state variables as well. This means that we need additional sensors for each state or use observers to reconstruct the states.

Hence, physically it is easier to implement a lead-lag compensator, but state space models will give more precise results.